

IT7003: Computer Networking 2

PBL Project Report

Semester:	Semester 2, 2025-2026
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Learning Outcomes Covered	The following learning outcomes will be tested: 1. Demonstrate advanced knowledge of the required protocols for VLAN implementation. 2. Implement and troubleshoot the VLAN Trunking Protocol (VTP), Inter-VLAN Routing and Spanning Tree Protocol (STP). 3. Demonstrate advanced knowledge of the required protocols for WAN implementation 4. Plan and configure LAN and WAN network connections and differentiate between WAN and LAN equipment and protocols. 5. Design a network security policy by using Access Control Lists (ACLs) and IP addressing services to implement this policy.
Weighting:	20% of the overall mark
Instructions	• Implement in groups of 4. • Write a report and create a Packet Tracer file on this network setup. • The report should document how you implemented each task. • Packet Tracer file should include a fully configured network. • Submit both the report and the implemented network on May 24, 2026, before 11:55pm. • If an assessment is submitted late, the maximum result the student can achieve is 60% in that assessment. • The cut off time for submitting an assessment will be 3 calendar days after the assessment is due. Submitting after 3 calendar days will get 0%. • Submission via Moodle. • Only one student in the group should submit the assessment. • Write all the device's password as a note in the packet tracer. • Each student is required to clearly indicate their name (inside the Packet Tracer) next to the branch they are responsible for configuring.

Scenario

Gulf Unified Financial Group (GUFG) has unified four previously independent financial companies under a centralized enterprise infrastructure. Each site historically operates separate networks, IP schemes, and services, limiting operational integration and centralized management. The new network design establishes interconnection among all locations while maintaining local functionality and centralized services. The architecture emphasizes scalability, redundancy, availability, and security to support enterprise operations. Implementation involves deploying network devices at each site and integrating them with the central hub, ensuring secure communications and high system availability through redundancy mechanisms. Future development will focus on enhancing automation, cloud integration, and advanced monitoring to optimize network performance and scalability.

Project Overview:

The project involves designing and implementing a unified enterprise network for multiple financial sites under centralized management. Each student is responsible for one site and must design IP addressing, routing, and security policies. The network must support inter-site connectivity and redundancy. WAN links will connect sites using dual authentication methods and failover mechanisms. The design will be tested in a simulation environment before real-world deployment.

Project Debrief

The project consists of two main parts, part 1 is the network planning and design report while part 2 is the implementation and testing of the design in simulation environment.

Part1: Network Planning and Design Report:

Your team is required to provide a detailed report on the network plan. Each student must handle one site. The report should include:

Task 1: IP Addressing & Routing Protocol Requirements

Design a structured private IPv4 addressing scheme will be implemented to ensure no IP conflicts and efficient routing across sites. Each site will use a unique subnet based on private address ranges, each site will use different routing protocols (OSPF, EIGRP, RIP v2, etc.), and route redistribution will be configured at the hub router to enable inter-protocol communication while preventing routing loops.

Give details on how this is achieved in the report for at all sites

SITE	ADDRESS RANGE	ROUTING PROTOCOL
BAHRAIN	10.0.0.0/16	OSPF
KSA	172.16.0.0/12	EIGRP AS 10
OMAN	192.168.0.0/16	EIGRP AS 20
KUWAIT	172.20.0.0/16	RIPv2

(CILO 1) (6Marks)

Task 2: Sites and Organizational Structure

To improve LAN performance and security, the network will be divided into multiple VLANs, separating users based on department type.

- Implementing VLANs on switches
- Assigning each department to its own VLAN
- Using inter-VLAN routing using Layer 3 Switches for controlled communication
- Reducing unnecessary broadcast traffic
- Give details on how this is achieved in the report and illustrate the performance improvement on the network at all sites when users of different types are separated.

Site A – Bahrain, Manama (Hup)

Total Employees: **260**

DEPARTMENT	EMPLOYEES
MANAGEMENT	25
ADMINISTRATION	40
FINANCE	50
IT	35
HUMAN RESOURCES	20
RISK & COMPLIANCE	25
MARKETING	30
CUSTOMER OPERATIONS	35

Site B – KSA, Riyadh

Total Employees: **170**

DEPARTMENT	EMPLOYEES
MANAGEMENT	15
ADMINISTRATION	25
FINANCE	30
IT	20
HUMAN RESOURCES	15
MARKETING	20
LEGAL	15
LOAN SERVICES	30

Site C – Muscat, Oman

Total Employees: **185**

DEPARTMENT	EMPLOYEES
MANAGEMENT	20
ADMINISTRATION	30
FINANCE	35
IT	25
HUMAN RESOURCES	15
INVESTMENT	25
AUDIT	15
CUSTOMER SUPPORT	20

Site D – Doha, Qatar

Total Employees: **210**

DEPARTMENT	EMPLOYEES
MANAGEMENT	20
ADMINISTRATION	35
FINANCE	40
IT	30
HUMAN RESOURCES	20
MARKETING	25
TREASURY	20
LEGAL	20

(CILOs 1, 2) (10 Marks)

Task 3: Security and Redundancy Measures on the LAN

For each site Design security and redundancy measures on the LAN (4 measures per category).

Security Measures:

1. Port security
2. DHCP Snooping
3. SSH-only management
4. Disabled unused ports

redundancy Measures:

1. Spanning Tree Protocol (STP, MST, RPVST)
2. Redundant links
3. Backup switches
4. First hop redundancy protocol

Detail in the report on how these measures will be implemented on site-A location LAN.

(CILOs 1, 2, 5) (8 Marks)

Task 4: Secure Wireless Access

Each site needs secure wireless access in addition to fixed wired access. Detail how secure wireless access will be added to the LAN at site-B. WPA2/WPA3 Enterprise encryption, use Separate wireless VLANs

(CILO 4) (6 Marks)

Task 5: Inter-Site Connectivity

All Sites will be interconnected using WAN links with reliable connectivity.

. Each WAN link must use a different data-link authentication protocol:

- One link → PPP Configuring PAP Authentication
- One link → PPP Configuring CHAP Authentication

Failover must be implemented using:

- Floating static routes

The topology must resemble a hub-and-spoke dual redundant architecture.

Details in the report how inter site connectivity is implemented at site-C.

(CILOs 3, 4) (6 Marks)

Task 6: Internet Connectivity

to provide an Internet service to all sites Each site must connect to the Internet using a local Service Provider (ISP) in its own country:

- KSA → Etisalat
- Qatar → Ooredoo
- Oman → Omantel
- Bahrain → Batelco

Each site:

- Has its own Internet edge router
- Implement PAT for internal users
- Each must implement Static NAT for Web Server and Email server
- Ensure no NAT conflicts with inter-site routing

Give details in the report about internet connectivity design for site-D

Task 7: Inter-Department Communication Policy

There are a number of different types of users on the network. We wish to restrict certain network access rights for different users.

- The IT staff should be the only staff members who can access network devices (routers and switches) through SSH.
 - For security purposes, block all ICMP traffic toward web and email servers.
 - Same departments across different sites must be allowed to communicate
 - Cross-department communication between different departments across sites must be restricted unless explicitly permitted
 - Block Human and Financial Resources department staff from having communications with the Internet but allow access to all other parts of the network using the named ACL.
 - Using a named ACL allows the Internet to access the servers for email and Internet traffic only. Block access to all other parts of the network
- Detail how Inter-Department Communication Policy at all sites

(CILO 5) (6 marks)

Part 2: Implementation and Troubleshooting

Task 8: Assign IPv4 addresses to all interfaces according to your design in the report's task 1. Proper Login banners must be configured on all key network devices and should only include necessary information.

(CILOs 1, 2) (6 Marks)

Task 9: Implement the method chosen to separate users as described in the report's task 2.

(CILOs 2) (6 Marks)

Task 10: Implement the security and redundancy measures designed in the report's task 3.

(CILOs 2, 5) (6 Marks)

Task 11: Implement wireless access at all sites.

(CILOs 2, 4) (6 Marks)

Task 12: Implement your WAN design for the connection between sites as detailed in task 5 of your report.

(CILO 4) (6 Marks)

Task 13: Implement Static NAT/PAT at all sites as detailed in task 6 of your report. **(CILO 5) (6 Marks)**

Task 14: Configure all access lists as requested in task 7 of the report. **(CILO 5) (6 Marks)**

Task 15: Prepare a complete and well-structured project report, including:

- Cover page
- Table of contents
- Clear task organization
- Network diagrams and configurations
- Proper referencing and citation APA style
- Appendices (e.g., configurations, screenshots, API usage if any)

(10 Marks)